Roll No. | | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-4)

PHARMACEUTICAL ORGANIC CHEMISTRY-III

Subject Code : BP-401 T

M Code : 75843

Date of Examination : 26-05-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Write short notes on:**

   a) E and Z configuration

   b) D and L configuration

   c) Eclipsed and staggered conformations.

   d) Centre of symmetry.

   e) Partial asymmetric synthesis.

   f) Basicity of pyrrole *vs.* pyridine.

   g) Birch reduction

   h) Aromaticity of pyrrole

   i) Wolf kischner reduction

   j) Structure and uses of oxazole.

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SECTION - B

2. What are sequence rules? Comment upon their role in assigning the RS nomenclature to optically active compounds.

3. Write a detailed note on nomenclature and classification of various Heterocyclic compounds.

4. **Explain :**

   a) Beckmann Rearrangement

   b) Oppenauer oxidation

   c) Claisen Schmidt condensation.

SECTION - C

5. Explain the reduction reactions carried out by utilizing different types of metal hydrides.

6. Discuss in detail asymmetric synthesis.

7. Discuss in detail various elements of symmetry and their significance in determining optical activity.

8. Write down physical methods utilized to determine the configuration of geometrical isomers.

9. Give various reactions of thiophene.

10. What is racemic mixture? Give various methods utilized to resolve it.

11. Explain stereoisomerism in biphenyl compounds.

12. Write the structure and uses of purine and Isoquinoline.

13. Write a detailed note on chemistry of Quinoline.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

2025